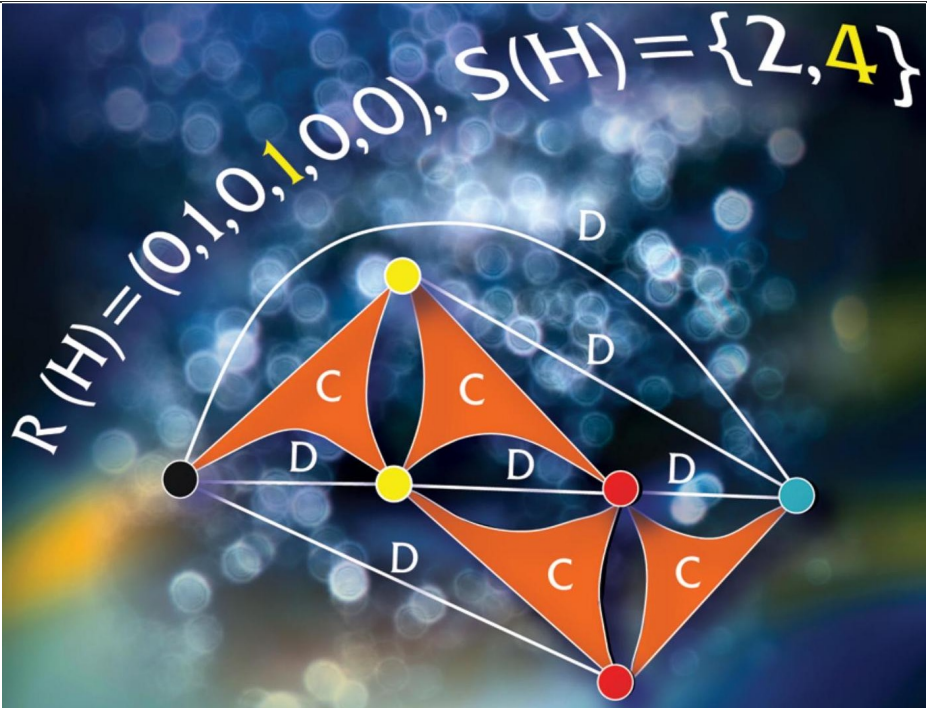
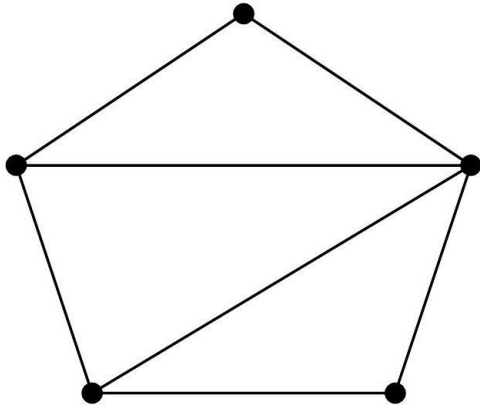
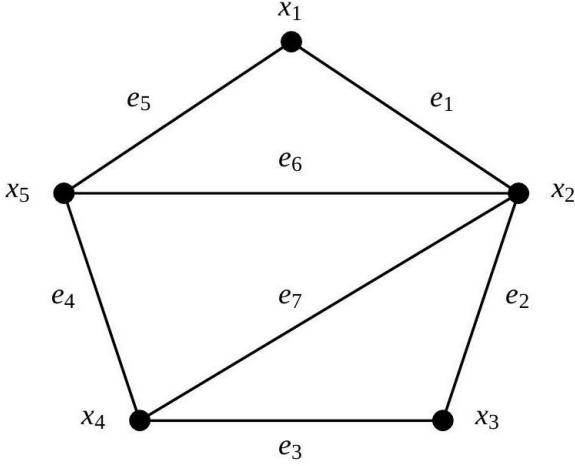
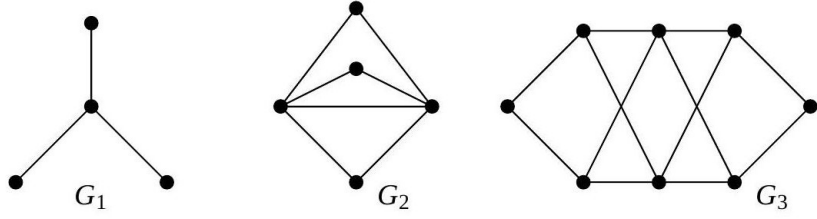
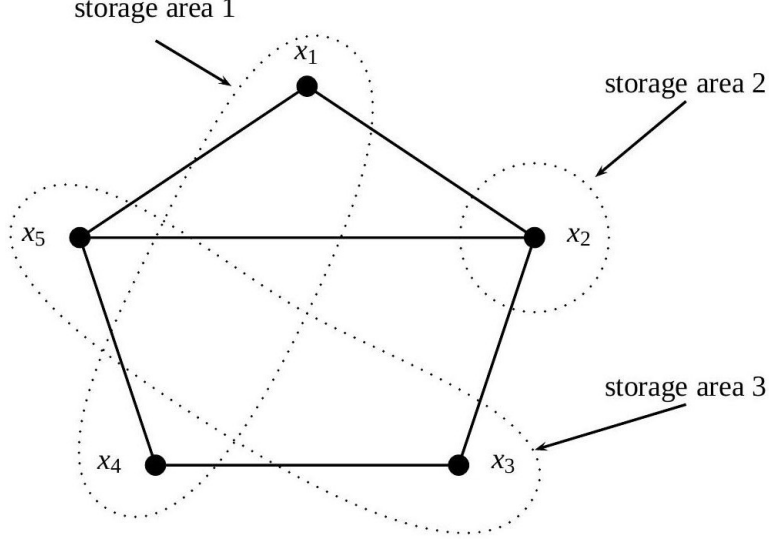
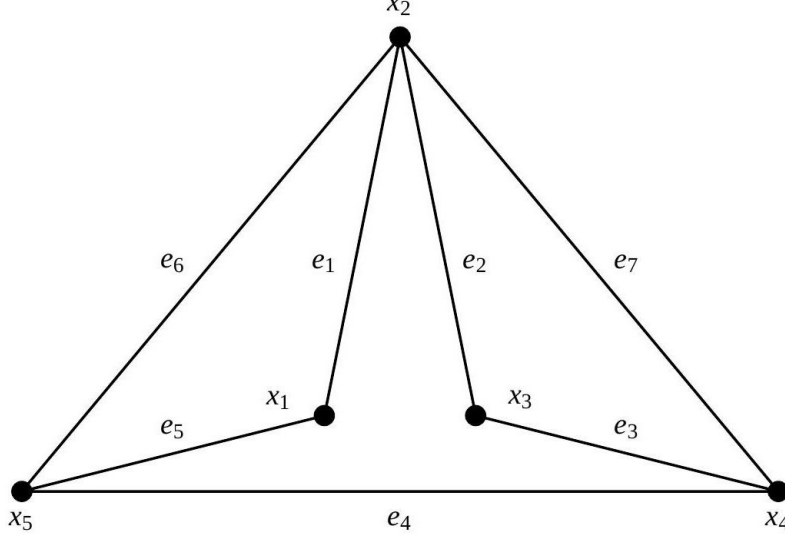
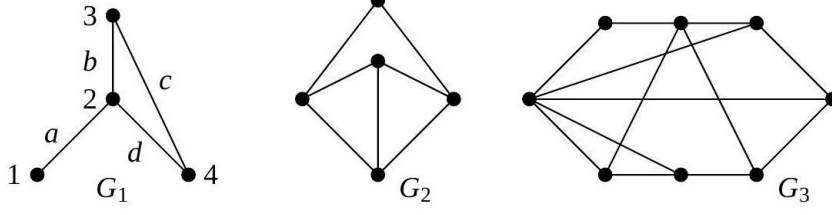
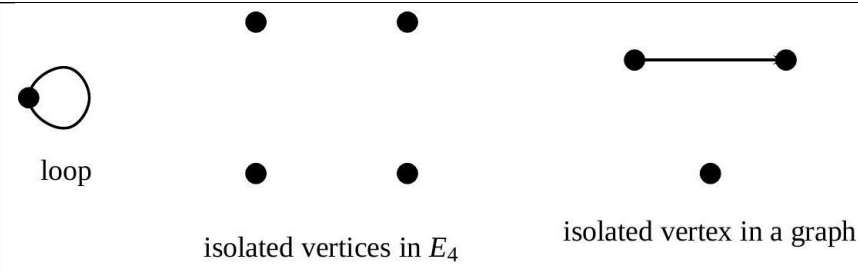
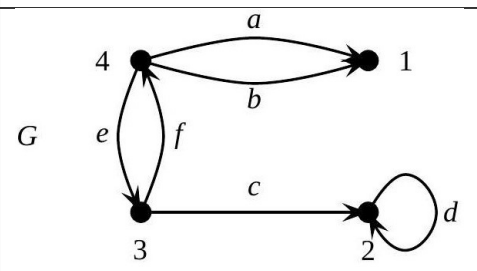
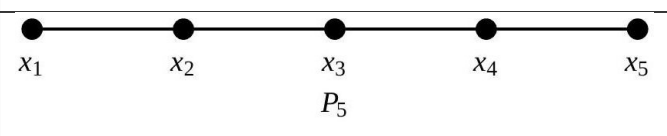
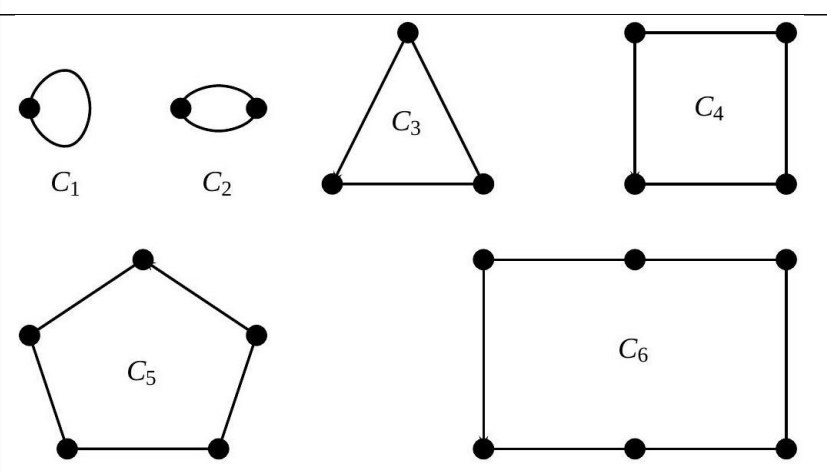
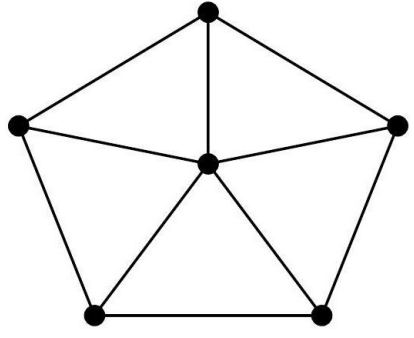
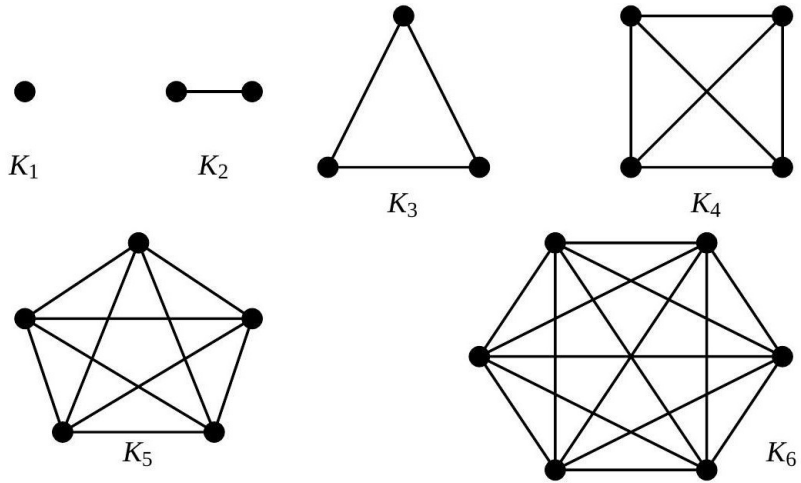
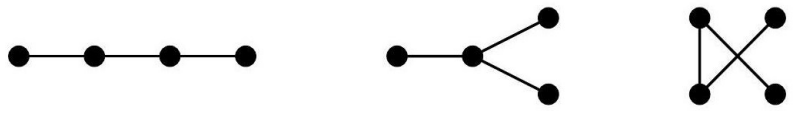
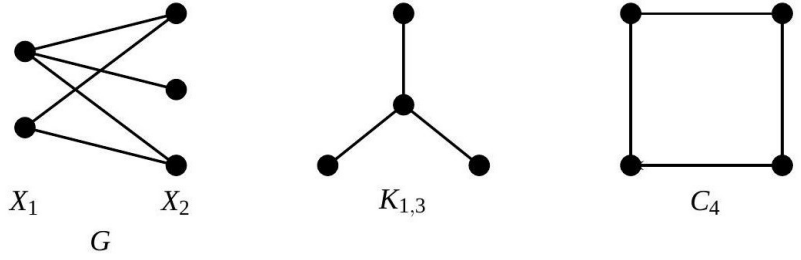
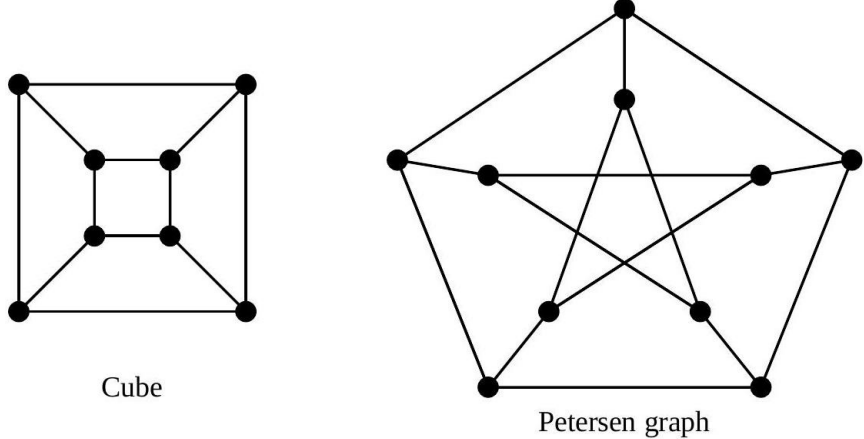


Image regions

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                 |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0001 | 1    | —     | —            | —        |   |
| voloshin-hypergraph_<br>DIA_0002 | 21   | —     | —            | —        |  |
| voloshin-hypergraph_<br>DIA_0003 | 22   | —     | —            | —        |  |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                 |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0004 | 24   | —     | —            | —        |    |
| voloshin-hypergraph_<br>DIA_0005 | 25   | —     | —            | —        |    |
| voloshin-hypergraph_<br>DIA_0006 | 29   | —     | —            | —        |   |
| voloshin-hypergraph_<br>DIA_0007 | 31   | —     | —            | —        |  |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                 |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0008 | 32   | —     | —            | —        |  <p>loop</p> <p>isolated vertices in <math>E_4</math></p> <p>isolated vertex in a graph</p> <p>multiple edges</p> <p>arcs</p>                      |
| voloshin-hypergraph_<br>DIA_0009 | 32   | —     | —            | —        |                                                                                                                                                    |
| voloshin-hypergraph_<br>DIA_0010 | 33   | —     | —            | —        |  <p><math>G</math></p>                                                                                                                             |
| voloshin-hypergraph_<br>DIA_0011 | 34   | —     | —            | —        |  <p><math>P_5</math></p>                                                                                                                          |
| voloshin-hypergraph_<br>DIA_0012 | 34   | —     | —            | —        |  <p><math>G_1</math></p> <p><math>G_2</math></p>                                                                                                 |
| voloshin-hypergraph_<br>DIA_0013 | 35   | —     | —            | —        |  <p><math>C_1</math></p> <p><math>C_2</math></p> <p><math>C_3</math></p> <p><math>C_4</math></p> <p><math>C_5</math></p> <p><math>C_6</math></p> |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                 |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0014 | 35   | —     | —            | —        |    |
| voloshin-hypergraph_<br>DIA_0015 | 36   | —     | —            | —        |    |
| voloshin-hypergraph_<br>DIA_0016 | 36   | —     | —            | —        |  |
| voloshin-hypergraph_<br>DIA_0017 | 37   | —     | —            | —        |  |
| voloshin-hypergraph_<br>DIA_0018 | 38   | —     | —            | —        |  |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0019 | 38   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0020 | 39   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0021 | 40   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0022 | 41   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0023 | 42   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0024 | 42   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0025 | 43   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0026 | 43   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0027 | 44   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0028 | 44   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0029 | 46   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0030 | 47   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0031 | 48   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0032 | 49   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0033 | 51   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0034 | 51   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0035 | 52   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0036 | 52   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0037 | 53   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0038 | 56   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0039 | 58   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0040 | 59   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0041 | 60   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0042 | 61   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|------|-------|--------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0043 | 64   | —     | —            | —        | <p>The diagram illustrates a relationship between two sets, <math>X</math> and <math>Y</math>. Set <math>X</math> is on the left and contains two subsets: <math>S</math> and <math>X - S_0</math>. Set <math>Y</math> is on the right and contains two subsets: <math>N_G(S)</math> and <math>N_G(S_0)</math>. Two edges, <math>G_A</math> and <math>G_B</math>, are shown. Edge <math>G_A</math> connects the subset <math>S</math> in <math>X</math> to the subset <math>N_G(S_0)</math> in <math>Y</math>. Edge <math>G_B</math> connects the subset <math>X - S_0</math> in <math>X</math> to the subset <math>N_G(S)</math> in <math>Y</math>.</p> |
| voloshin-hypergraph_<br>DIA_0044 | 65   | —     | —            | —        | <p>The diagram shows two sets, <math>X</math> and <math>Y</math>. Set <math>X</math> contains subsets <math>S</math> and <math>A</math>. Set <math>Y</math> contains subsets <math>N_{G_1}(S)</math> and <math>B</math>. A horizontal dashed line divides the diagram into two regions: <math>G_1</math> above and <math>G_2</math> below. Two edges are shown: one connecting <math>S</math> to <math>N_{G_1}(S)</math> and another connecting <math>A</math> to <math>B</math>.</p>                                                                                                                                                                    |
| voloshin-hypergraph_<br>DIA_0045 | 66   | —     | —            | —        | <p>The diagram depicts a grid of points arranged in four rows and six columns. Horizontal and vertical lines connect the points, forming a series of adjacent rectangles. This structure represents a hypergraph or a specific graph configuration.</p>                                                                                                                                                                                                                                                                                                                                                                                                  |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0046 | 68   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0047 | 68   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0048 | 70   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0049 | 72   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0050 | 74   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0051 | 75   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0052 | 75   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0053 | 76   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0054 | 76   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0055 | 77   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0056 | 78   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0057 | 79   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0058 | 80   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0059 | 81   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0060 | 81   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0061 | 84   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0062 | 84   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0063 | 86   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0064 | 87   | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0065 | 88   | —     | —            | —        | <p>The diagram illustrates the construction of a hypergraph from two base graphs, <math>K_5</math> and <math>K_{3,3}</math>.</p> <p><b>Top section (<math>K_5</math>):</b> A complete graph <math>K_5</math> with vertices 1, 2, 3, 4, 5 is shown. It is transformed into a triangle with vertices 1, 2, 3. Vertex 4 is placed either 'inside' or 'outside' the triangle. In both cases, the resulting graph has 'no face for 5'.</p> <p><b>Bottom section (<math>K_{3,3}</math>):</b> A complete bipartite graph <math>K_{3,3}</math> with vertices 1, 2, 3, 4, 5, 6 is shown. It is transformed into a square with vertices 1, 2, 3, 4. Vertex 5 is placed either 'inside' or 'outside' the square. In both cases, the resulting graph has 'no face for 6'.</p> |
| voloshin-hypergraph_<br>DIA_0066 | 91   | —     | —            | —        | <p>Two graphs, <math>G_1</math> and <math>G_2</math>, are shown. <math>G_1</math> is a complex graph with 10 vertices and many edges. <math>G_2</math> is a graph with 10 vertices and 12 edges, including a cycle of length 6.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| voloshin-hypergraph_<br>DIA_0067 | 92   | —     | —            | —        | <p>A graph with 6 vertices and 10 edges, including a cycle of length 6.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0068 | 93   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0069 | 97   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0070 | 99   | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0071 | 100  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------|------|-------|--------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0072 | 101  | —     | —            | —        | <p>Diagram illustrating the decomposition of a hypergraph <math>G</math> into two hypergraphs <math>G_1</math> and <math>G_2</math>. <math>G</math> consists of two vertices <math>x</math> and <math>y</math> connected to a set of three vertices. <math>G_1</math> is identical to <math>G</math> but with a dashed edge between <math>x</math> and <math>y</math>. <math>G_2</math> consists of a single vertex <math>xy</math> connected to the same set of three vertices. The equation <math>G = G_1 + G_2</math> is shown.</p> |
| voloshin-hypergraph_<br>DIA_0073 | 102  | —     | —            | —        | <p>Diagram illustrating the decomposition of a tree <math>G</math> into two subgraphs <math>G_1</math> and <math>G_2</math>. <math>G</math> is a tree with vertices <math>x</math>, <math>y</math>, and <math>z</math>. <math>G_1 = K_3</math> is a triangle with vertices <math>x</math>, <math>y</math>, and <math>z</math>. <math>G_2 = K_2</math> is a path with vertices <math>y</math> and <math>xz</math>. The equation <math>G = G_1 + G_2</math> is shown.</p>                                                                |
| voloshin-hypergraph_<br>DIA_0074 | 103  | —     | —            | —        | <p>Diagram illustrating a hierarchical tree structure <math>G</math>. The root node <math>G</math> branches into <math>G_1</math> and <math>G_2</math>. <math>G_1</math> branches into <math>G_3</math> and <math>G_4</math>, and <math>G_2</math> branches into <math>G_5</math> and <math>G_6</math>. Below these, a sequence of nodes <math>K_{i_1}, K_{i_2}, \dots, K_{i_s}</math> is shown, connected by a dotted line.</p>                                                                                                       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0075 | 104  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0076 | 105  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0077 | 107  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0078 | 108  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0079 | 109  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0080 | 111  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0081 | 112  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0082 | 113  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0083 | 115  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0084 | 117  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0085 | 119  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0086 | 120  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0087 | 121  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0088 | 122  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0089 | 124  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0090 | 125  | —     | —            | —        | <p>The diagram shows a transformation. On the left, a single vertical edge connects two vertices. An arrow points to the right, where a more complex graph is shown. This graph has five vertices. It consists of a triangle on the right, a diamond shape in the middle, and a vertical edge on the left. A dotted ellipse encircles the two vertices of the diamond that are closest to the triangle.</p>                                                                                                                               |
| voloshin-hypergraph_<br>DIA_0091 | 125  | —     | —            | —        | <p>A complex graph with 10 vertices. It features a central vertex connected to several other vertices, forming a series of interconnected triangles and quadrilaterals. The overall shape is somewhat circular with a central hub.</p>                                                                                                                                                                                                                                                                                                    |
| voloshin-hypergraph_<br>DIA_0092 | 126  | —     | —            | —        | <p>Two graphs are shown side-by-side. The left graph, labeled <math>G</math>, has five vertices labeled 1, 2, 3, 2, 3. The vertices are arranged in a pentagonal shape with internal edges. The right graph, labeled <math>\bar{G}</math>, has five vertices arranged in a similar pentagonal shape but with different internal connections. Both graphs have dotted ellipses around some of their vertices.</p>                                                                                                                          |
| voloshin-hypergraph_<br>DIA_0093 | 129  | —     | —            | —        | <p>Two transformations are shown. The top transformation shows a graph <math>G</math> on the left with vertices labeled <math>a, b, c, d, e</math> and an arrow pointing to a graph <math>L(G)</math> on the right with vertices labeled <math>a, b, c, d, e</math>. The bottom transformation shows a graph on the left with vertices labeled 1, 2, 3, 4 and an arrow pointing to a graph on the right with vertices labeled 1, 2, 3, 4. The left graph is labeled 'edge coloring' and the right graph is labeled 'vertex coloring'.</p> |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0094 | 130  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0095 | 131  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0096 | 133  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0097 | 134  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                 |
|----------------------------------|------|-------|--------------|----------|-----------------------|
| voloshin-hypergraph_<br>DIA_0098 | 136  | —     | —            | —        | <p><math>G</math></p> |
| voloshin-hypergraph_<br>DIA_0099 | 140  | —     | —            | —        |                       |
| voloshin-hypergraph_<br>DIA_0100 | 141  | —     | —            | —        |                       |

| Identifier                                                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div>voloshin-hypergraph_</div> <div>DIA_0101</div> </div> | 145  | —     | —            | —        | <p>The image displays three network flow diagrams arranged vertically, connected by downward arrows, illustrating the progression of a flow augmentation algorithm. Each diagram features a source node <math>u</math> and a sink node <math>v</math>, with intermediate nodes <math>x_1, x_2, x_3, x_4</math>.</p> <ul style="list-style-type: none"> <li><b>Initial flow <math>F_0</math>:</b> <ul style="list-style-type: none"> <li>Edges and flows: <math>u \rightarrow x_1</math> (0,3), <math>u \rightarrow x_4</math> (1,5), <math>x_1 \rightarrow x_2</math> (1,7), <math>x_1 \rightarrow x_3</math> (1,1), <math>x_2 \rightarrow v</math> (1,6), <math>x_3 \rightarrow v</math> (0,2), <math>x_4 \rightarrow x_3</math> (1,4).</li> </ul> </li> <li><b>Flow <math>F_1</math>:</b> <ul style="list-style-type: none"> <li>Edges and flows: <math>u \rightarrow x_1</math> (1,3), <math>u \rightarrow x_4</math> (1,5), <math>x_1 \rightarrow x_2</math> (1,7), <math>x_1 \rightarrow x_3</math> (0,1), <math>x_2 \rightarrow v</math> (1,6), <math>x_3 \rightarrow v</math> (1,2), <math>x_4 \rightarrow x_3</math> (1,4).</li> </ul> </li> <li><b>Flow <math>F_2</math>:</b> <ul style="list-style-type: none"> <li>Edges and flows: <math>u \rightarrow x_1</math> (3,3), <math>u \rightarrow x_4</math> (1,5), <math>x_1 \rightarrow x_2</math> (3,7), <math>x_1 \rightarrow x_3</math> (0,1), <math>x_2 \rightarrow v</math> (3,6), <math>x_3 \rightarrow v</math> (1,2), <math>x_4 \rightarrow x_3</math> (1,4).</li> </ul> </li> </ul> |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                           |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0102 | 146  | —     | —            | —        | <p>The top diagram shows a flow network with source <math>u</math> and sink <math>v</math>. Intermediate nodes are <math>x_1, x_2, x_3, x_4</math>. The flow <math>F_3</math> is indicated. The bottom diagram shows the same network with a maximum flow <math>F_4</math>.</p> |
| voloshin-hypergraph_<br>DIA_0103 | 151  | —     | —            | —        | <p>A hypergraph diagram with 6 nodes and 4 hyperedges. The nodes are arranged in a hexagonal pattern, and the hyperedges are represented by closed curves connecting different subsets of nodes.</p>                                                                            |
| voloshin-hypergraph_<br>DIA_0104 | 155  | —     | —            | —        | <p>Two hypergraph diagrams labeled <math>\mathcal{H}_1</math> and <math>\mathcal{H}_2</math>. <math>\mathcal{H}_1</math> is a smaller hypergraph with 6 nodes and 4 hyperedges. <math>\mathcal{H}_2</math> is a larger hypergraph with 8 nodes and 6 hyperedges.</p>            |

| Identifier                       | Page  | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----------------------------------|-------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|-------|-------|-------|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| voloshin-hypergraph_<br>DIA_0105 | 157   | —     | —            | —        | <p>The diagram illustrates the relationship between a hypergraph <math>\mathcal{H} = (X, \mathcal{D})</math>, its incidence matrix <math>I(\mathcal{H})</math>, and its bipartite graph <math>B(\mathcal{H})</math>.</p> <p><b>Hypergraph <math>\mathcal{H}</math>:</b> A set of 6 vertices <math>\{1, 2, 3, 4, 5, 6\}</math> and 5 hyperedges <math>\{D_1, D_2, D_3, D_4, D_5\}</math>. Vertex 1 is in <math>D_1</math>, vertex 2 is in <math>D_2</math>, vertex 3 is in <math>D_4</math>, vertex 4 is in <math>D_5</math>, and vertex 5 is in <math>D_3</math>. Vertex 6 is not in any hyperedge.</p> <p><b>Incidence Matrix <math>I(\mathcal{H})</math>:</b></p> <table border="1"> <thead> <tr> <th></th><th><math>D_1</math></th><th><math>D_2</math></th><th><math>D_3</math></th><th><math>D_4</math></th><th><math>D_5</math></th></tr> </thead> <tbody> <tr> <th>1</th><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr> <tr> <th>2</th><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr> <tr> <th>3</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>4</th><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr> <th>5</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>6</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </tbody> </table> <p><b>Bipartite Graph <math>B(\mathcal{H})</math>:</b> A bipartite graph with vertices <math>\{1, 2, 3, 4, 5, 6\}</math> on the left and <math>\{D_1, D_2, D_3, D_4, D_5\}</math> on the right. Edges connect vertices to the hyperedges they belong to: <math>(1, D_1), (2, D_2), (3, D_4), (4, D_5), (5, D_3)</math>.</p> <p><b>Hypergraph <math>\mathcal{H}^*</math>:</b> A set of 6 vertices <math>\{1, 2, 3, 4, 5, 6\}</math> and 5 hyperedges <math>\{d_1, d_2, d_3, d_4, d_5\}</math>. Vertex 1 is in <math>d_1</math>, vertex 2 is in <math>d_2</math>, vertex 3 is in <math>d_3</math>, vertex 4 is in <math>d_4</math>, and vertex 5 is in <math>d_5</math>. Vertex 6 is not in any hyperedge.</p> |  | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | 1 | 1 | 1 | 1 | 0 | 0 | 2 | 0 | 1 | 1 | 1 | 0 | 3 | 0 | 0 | 0 | 1 | 1 | 4 | 0 | 0 | 1 | 0 | 1 | 5 | 0 | 0 | 0 | 1 | 1 | 6 | 0 | 0 | 0 | 0 | 0 |
|                                  | $D_1$ | $D_2$ | $D_3$        | $D_4$    | $D_5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                | 1     | 1     | 1            | 0        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2                                | 0     | 1     | 1            | 1        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                | 0     | 0     | 0            | 1        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                | 0     | 0     | 1            | 0        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5                                | 0     | 0     | 0            | 1        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 6                                | 0     | 0     | 0            | 0        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| voloshin-hypergraph_<br>DIA_0106 | 158   | —     | —            | —        | <p>The diagram illustrates the relationship between a graph <math>G</math> and its dual graph <math>G^*</math>.</p> <p><b>Graph <math>G</math>:</b> A graph with 5 vertices <math>\{1, 2, 3, 4, 5\}</math> and 5 edges <math>\{a, b, c, d, e\}</math>. Vertex 1 is connected to 2 by edge <math>a</math>, vertex 2 is connected to 3 by edge <math>b</math>, vertex 3 is connected to 4 by edge <math>c</math>, vertex 4 is connected to 5 by edge <math>d</math>, and vertex 5 is connected to 1 by edge <math>e</math>.</p> <p><b>Dual Graph <math>G^*</math>:</b> A graph with 5 vertices <math>\{a, b, c, d, e\}</math> and 5 edges <math>\{1, 2, 3, 4, 5\}</math>. Vertex <math>a</math> is connected to <math>b</math> by edge 1, vertex <math>b</math> is connected to <math>c</math> by edge 2, vertex <math>c</math> is connected to <math>d</math> by edge 3, vertex <math>d</math> is connected to <math>e</math> by edge 4, and vertex <math>e</math> is connected to <math>a</math> by edge 5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |       |       |       |       |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0107 | 161  | —     | —            | —        | <p>Diagrams illustrating various hypergraph types <math>K_4^i</math> for <math>i = 0, 1, 2, 3, 4</math>. <math>K_4^0</math> is a single vertex. <math>K_4^1</math> is a single edge (two vertices). <math>K_4^2</math> is a square with both diagonals. <math>K_4^3</math> is a complete graph <math>K_4</math> with four vertices and six edges. <math>K_4^4</math> is a single hyperedge containing four vertices.</p>                              |
| voloshin-hypergraph_<br>DIA_0108 | 163  | —     | —            | —        | <p>Diagram illustrating the strong deletion of a vertex <math>x</math> from a hypergraph <math>\mathcal{H}</math>. The left side shows <math>\mathcal{H}</math> with vertex <math>x</math> (circled) and its incident edges. An arrow labeled "delete <math>x</math> strongly" points to the resulting hypergraph <math>\mathcal{H}_1 = \mathcal{H} - x</math>, where <math>x</math> and its incident edges are removed.</p>                          |
| voloshin-hypergraph_<br>DIA_0109 | 163  | —     | —            | —        | <p>Diagram illustrating the weak deletion of a vertex <math>x</math> from a hypergraph <math>\mathcal{H}</math>. The left side shows <math>\mathcal{H}</math> with vertex <math>x</math> (circled) and its incident edges. An arrow labeled "delete <math>x</math> weakly" points to the resulting hypergraph <math>\mathcal{H}_1 = \mathcal{H} - x</math>, where <math>x</math> is removed but its incident edges remain as separate hyperedges.</p> |
| voloshin-hypergraph_<br>DIA_0110 | 164  | —     | —            | —        | <p>Diagram illustrating the strong deletion of a set <math>D</math> from a hypergraph <math>\mathcal{H}</math>. The left side shows <math>\mathcal{H}</math> with set <math>D</math> (circled) and its incident edges. An arrow labeled "delete <math>D</math> strongly" points to the resulting hypergraph <math>\mathcal{H}_1 = \mathcal{H} - D</math>, where <math>D</math> and its incident edges are removed.</p>                                |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0111 | 164  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0112 | 165  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0113 | 166  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0114 | 168  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0115 | 171  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0116 | 172  | —     | —            | —        |       |

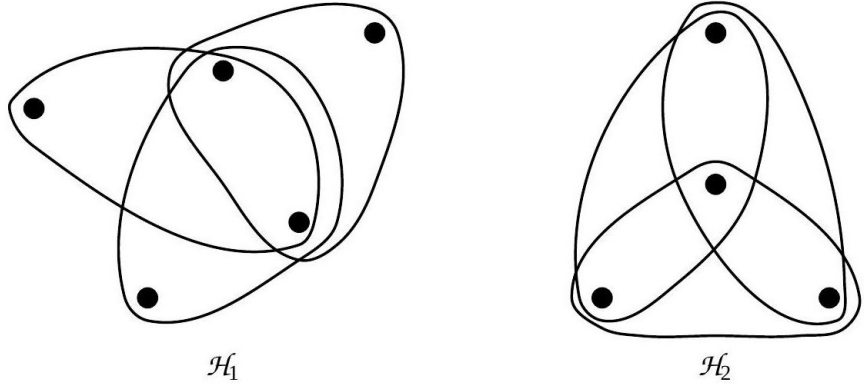
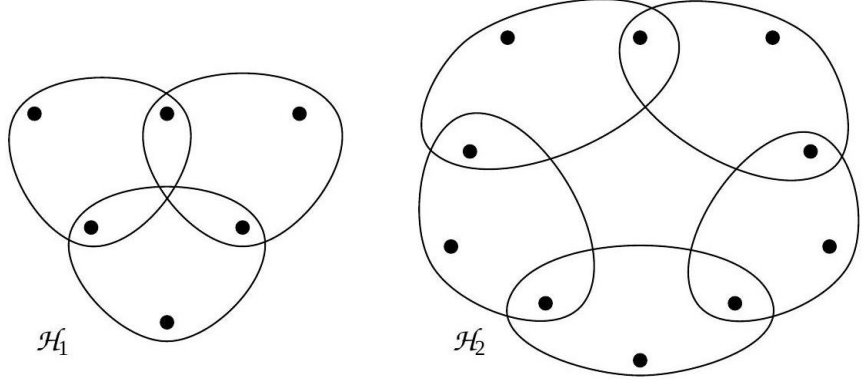
| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                               |
|----------------------------------|------|-------|--------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0117 | 174  | —     | —            | —        | <p> <math>\mathcal{H} = K_3</math><br/> not conformal, not Helly </p> <p> <math>\mathcal{H}_1</math> - conformal, not Helly </p> <p> <math>\mathcal{H}_2 \cong \mathcal{H}_1^*</math> - Helly,<br/> not conformal </p> <p> <math>\mathcal{H}_3 \cong \mathcal{H}_3^*</math> - Helly, conformal </p> |
| voloshin-hypergraph_<br>DIA_0118 | 178  | —     | —            | —        | <p> <math>T</math> </p> <p> <math>C_4</math> </p> <p> <math>T</math> </p> <p> <math>C_4</math> </p>                                                                                                                                                                                                 |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                               |
|----------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0119 | 179  | —     | —            | —        | <p><math>\mathcal{H}_1</math>      <math>\mathcal{H}_2</math></p>                   |
| voloshin-hypergraph_<br>DIA_0120 | 180  | —     | —            | —        | <p><math>\mathcal{H}</math>      <math>\mathcal{H}_0</math></p>                     |
| voloshin-hypergraph_<br>DIA_0121 | 181  | —     | —            | —        | <p><math>\mathcal{H}_0^*</math>      <math>\mathcal{H}^*</math></p>                 |
| voloshin-hypergraph_<br>DIA_0122 | 181  | —     | —            | —        | <p><math>\mathcal{H}_0^*</math>      <math>\mathcal{H}^*</math></p>                 |
| voloshin-hypergraph_<br>DIA_0123 | 182  | —     | —            | —        | <p><math>G</math>      <math>\mathcal{H}</math>      <math>\mathcal{H}^*</math></p> |

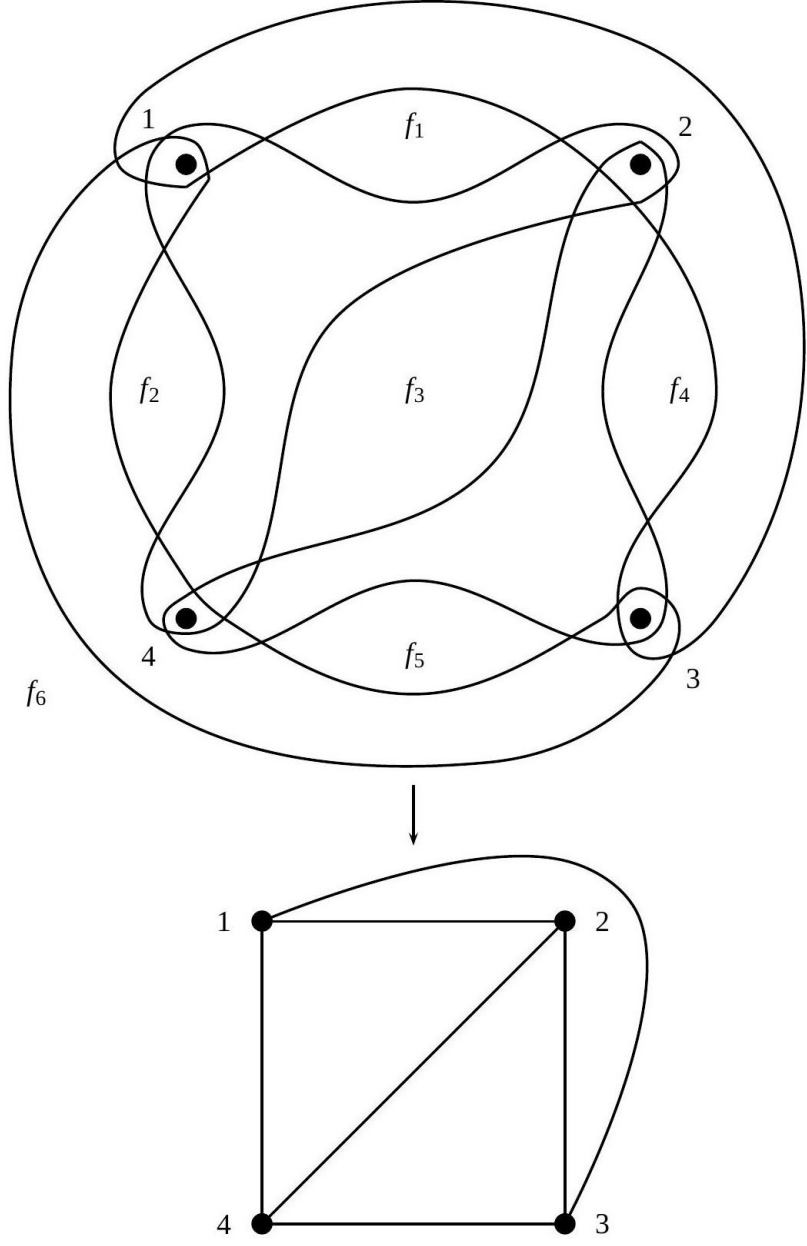
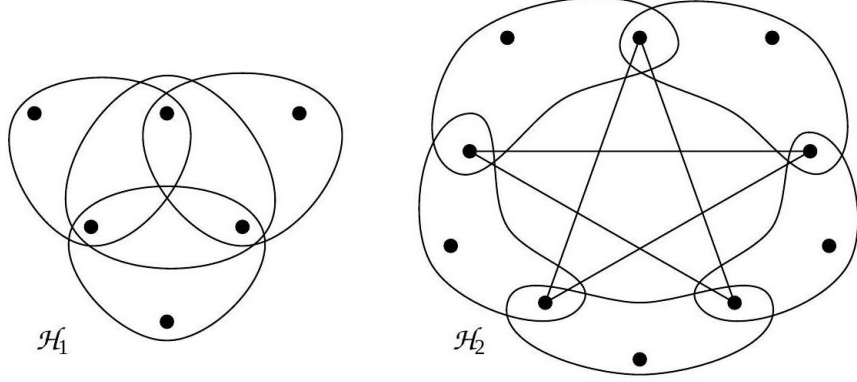
| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0124 | 183  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0125 | 186  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0126 | 187  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0127 | 188  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0128 | 189  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0129 | 191  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0130 | 194  | —     | —            | —        | <p>Diagram illustrating a hypergraph <math>\mathcal{H}</math> and its associated tree <math>T \subseteq [\mathcal{H}]_2</math>. The top part shows <math>\mathcal{H}</math> with a vertex <math>x</math> and an arrow pointing to <math>T</math>. The bottom part shows <math>\mathcal{H} - x</math> and an arrow pointing to <math>T - x</math>.</p> |
| voloshin-hypergraph_<br>DIA_0131 | 196  | —     | —            | —        | <p>Diagram illustrating two hypergraphs <math>\mathcal{H}_1</math> and <math>\mathcal{H}_2</math>.</p>                                                                                                                                                                                                                                                |
| voloshin-hypergraph_<br>DIA_0132 | 198  | —     | —            | —        | <p>Diagram illustrating a hypergraph structure with eight vertices and four hyperedges.</p>                                                                                                                                                                                                                                                           |
| voloshin-hypergraph_<br>DIA_0133 | 199  | —     | —            | —        | <p>Diagram illustrating two hypergraph structures.</p>                                                                                                                                                                                                                                                                                                |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                               |
|----------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0134 | 200  | —     | —            | —        |  |
| voloshin-hypergraph_<br>DIA_0135 | 202  | —     | —            | —        |  |

| Identifier                                                       | Page | Conf. | LaTeX source | Rendered | Image |
|------------------------------------------------------------------|------|-------|--------------|----------|-------|
| <div> <div>voloshin-hypergraph_</div> <div>DIA_0136</div> </div> | 203  | —     | —            | —        |       |

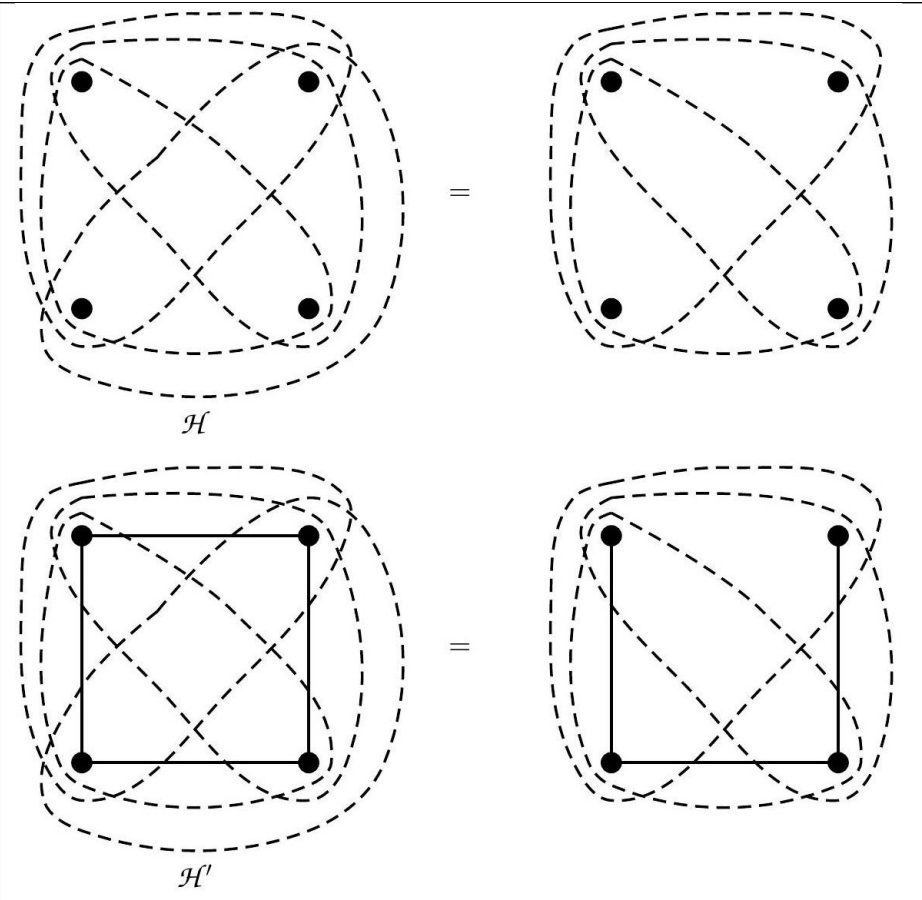
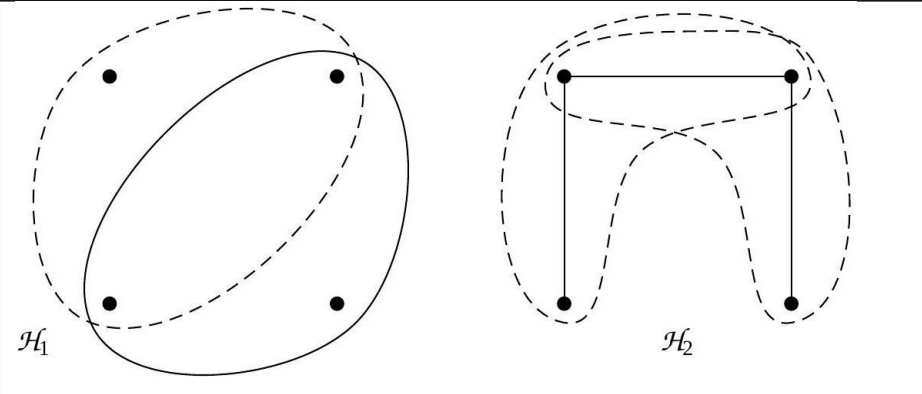
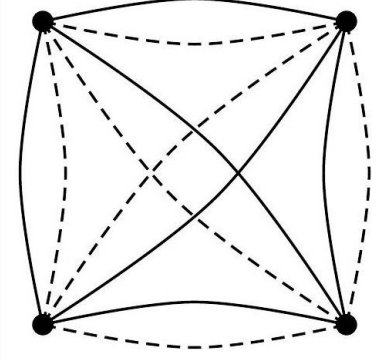
| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                 |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0137 | 206  | —     | —            | —        |   |
| voloshin-hypergraph_<br>DIA_0138 | 207  | —     | —            | —        |  |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0139 | 210  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0140 | 213  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0141 | 216  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0142 | 217  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0143 | 218  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0144 | 221  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0145 | 222  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0146 | 223  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0147 | 226  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0148 | 229  | —     | —            | —        |       |

| Identifier                                                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------|------|-------|--------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div>voloshin-hypergraph_</div> <div>DIA_0149</div> </div> | 232  | —     | —            | —        | <p>The diagram illustrates a sequence of operations on a hypergraph <math>C</math> (represented by a diamond shape with nodes 1, 2, 3, 4 and edges <math>D_1, D_2, D_3, D_4</math>):</p> <ul style="list-style-type: none"> <li><b>Top row:</b> <math>C</math> is shown as a sum of two diamond shapes. The first diamond has edges <math>D_1, D_2, D_3, D_4</math>. The second diamond has edges <math>D_1, D_2, D_3</math> and a dashed edge <math>D_4</math>. This is labeled "splitting 13".</li> <li><b>Second row:</b> The first diamond is further split into two diamond shapes. The second diamond is split into a triangle (nodes 1, 2, 3) and a node 4. This is labeled "spl. 24".</li> <li><b>Third row:</b> The first diamond is further split into two diamond shapes. The second diamond is split into a triangle (nodes 1, 2, 3) and a node 4. This is labeled "C-clearing + contraction 13".</li> <li><b>Fourth row:</b> The first diamond is further split into two diamond shapes. The second diamond is split into a triangle (nodes 1, 2, 3) and a node 4. This is labeled "elimination".</li> <li><b>Fifth row:</b> The first diamond is further split into two diamond shapes. The second diamond is split into a triangle (nodes 1, 2, 3) and a node 4. This is labeled "C-clearing + contraction 24".</li> <li><b>Sixth row:</b> The first diamond is further split into two diamond shapes. The second diamond is split into a triangle (nodes 1, 2, 3) and a node 4. This is labeled "contraction 24".</li> <li><b>Seventh row:</b> The final simplified state, showing a triangle (nodes 1, 2, 3) and a node 4, and a node 13.</li> </ul> |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                 |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0150 | 234  | —     | —            | —        |    |
| voloshin-hypergraph_<br>DIA_0151 | 235  | —     | —            | —        |  |
| voloshin-hypergraph_<br>DIA_0152 | 237  | —     | —            | —        |  |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image         |
|----------------------------------|------|-------|--------------|----------|---------------|
| voloshin-hypergraph_<br>DIA_0153 | 238  | —     | —            | —        |               |
| voloshin-hypergraph_<br>DIA_0154 | 239  | —     | —            | —        |               |
| voloshin-hypergraph_<br>DIA_0155 | 242  | —     | —            | —        |               |
| voloshin-hypergraph_<br>DIA_0156 | 242  | —     | —            | —        | $\mathcal{H}$ |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0157 | 244  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0158 | 250  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0159 | 251  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0160 | 253  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0161 | 257  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0162 | 258  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0163 | 259  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0164 | 261  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0165 | 262  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0166 | 265  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0167 | 267  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0168 | 269  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                   |
|----------------------------------|------|-------|--------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0169 | 271  | —     | —            | —        |                                                                                                                                                                                                                                                                                                         |
| voloshin-hypergraph_<br>DIA_0170 | 275  | —     | —            | —        | <p>1st coloring: <math>ABBA</math></p> <p>2nd coloring: <math>ABAB</math></p> <p>3rd coloring: <math>AABB</math></p> <p><math>\Rightarrow R(\mathcal{H}) = (0, 3, 0, 0), P(\mathcal{H}, \lambda) = 3\lambda(\lambda - 1)</math></p> <p><math>\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2</math></p> |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image |
|----------------------------------|------|-------|--------------|----------|-------|
| voloshin-hypergraph_<br>DIA_0171 | 276  | —     | —            | —        |       |
| voloshin-hypergraph_<br>DIA_0172 | 277  | —     | —            | —        |       |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0173 | 280  | —     | —            | —        | <p>The diagram illustrates a mapping <math>G</math> from a graph to a hypergraph. The top graph has nodes <math>Z_1, Z_2, Z_3</math> and edges <math>\{a, c\}, \{a, b\}, \{b, c\}</math>. The bottom hypergraph <math>\mathcal{H} = (X, \mathcal{C}, \mathcal{D})</math> has nodes <math>a, b, c</math> and hyperedges <math>Z_1, Z_2, Z_3</math>. A double-headed arrow labeled <math>G</math> connects the two diagrams.</p>                                                      |
| voloshin-hypergraph_<br>DIA_0174 | 281  | —     | —            | —        | <p>The diagram illustrates a mapping <math>G</math> from a bipartite graph to a hypergraph. The top graph has nodes <math>x_1, x_2, x_3, x_4</math> (jobs) and <math>y_1, y_2, y_3, y_4, y_5</math> (resources) with various edges. The bottom hypergraph <math>\mathcal{H} = (X, \mathcal{C}, \mathcal{D})</math> has nodes <math>x_1, x_2, x_3, x_4</math> and hyperedges <math>Z_1, Z_2, Z_3</math>. A double-headed arrow labeled <math>G</math> connects the two diagrams.</p> |

| Identifier                       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                             |
|----------------------------------|------|-------|--------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| voloshin-hypergraph_<br>DIA_0175 | 284  | —     | —            | —        | <div> <div> <div>Induction basis</div> <div>Induction hypothesis</div> <div>Induction step</div> </div> <p>1. <math>P_1</math></p> <p>2. <math>P_k \rightarrow P_{k+1}</math></p> <p><math>\longrightarrow P_n</math> is true for all <math>n = 1, 2, \dots</math></p> </div>     |
| voloshin-hypergraph_<br>DIA_0176 | 284  | —     | —            | —        | <div> <div>Induction basis</div> <div>Induction hypothesis</div> <div>Induction step</div> </div> <p>1. <math>P_1</math></p> <p>2. <math>P_1, P_2, \dots, P_k \rightarrow P_{k+1}</math></p> <p><math>\longrightarrow P_n</math> is true for all <math>n = 1, 2, \dots</math></p> |